

Index-Based Price Adjustment in US Strategic Petroleum Reserve Drawdowns

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Abstract

Sales from the US Strategic Petroleum Reserve (SPR) are conducted using basis auctions, in which winning bidders’ final payments are indexed to spot prices. This mechanism can eliminate potential inefficiencies arising from private information about the evolution of oil prices between auction and delivery, benefiting the seller (the US government). Reduced form evidence suggests that bids in SPR sales are consistent with pure private values. In a stylized model calibrated to the 2022 SPR drawdown, the use of basis auctions plausibly generates tens of millions of dollars in additional revenue.

Keywords: Strategic petroleum reserve; contingent payment auctions; private information; oil markets

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1 Introduction

In 2022, the Department of Energy (DOE) auctioned 180 million barrels of crude oil from the United States Strategic Petroleum Reserve (SPR). A distinctive feature of the auction mechanism used by DOE is *price indexing*: a winning bidder’s final payment is linked to the value of an oil price index at delivery, weeks or months after the auction.

The auctions literature has identified two key mechanisms by which ex post price adjustment may benefit a seller. First, making payments contingent upon information revealed after the auction can reduce information rents (Hansen, 1985; Riley, 1988). Second, contingent payments can mitigate the winner’s curse: the harm from overestimating the value of a good is reduced if the final payment is adjusted downwards when it turns out to be less valuable (Skrzypacz, 2013). Whether these effects are large in a given setting depends on the structure of bidders’ valuations, the nature of bidders’ information, and the details of the auction protocol adopted by the seller.

This paper investigates the performance of DOE’s form of index-based price adjustment during the 2022 drawdown. I present empirical evidence consistent with large revenue gains. To motivate the empirical analysis, I first present a stylized model capturing the forces above. With a suitable price index, price indexing reduces the significance of private information about the common component of valuations, compressing the distribution of types and weakening adverse selection to the seller’s benefit. The benefits of price indexing are greatest when the price index exactly coincides with the common component of valuations, in which case adverse selection is eliminated altogether. In the next part of the paper, I test for violations of this condition in bidding data from the 2022 drawdown. I find no evidence of adverse selection after price indexing. The key question is therefore whether private information would have been a more significant factor had DOE used a different format, such as a fixed price auction. Absent direct evidence of private information, I conclude by showing that there exists a large range of plausible information structures for which price indexing could have generated tens of millions of dollars in additional revenue in 2022.

I begin by analyzing the impact of price indexing in the context of Goeree and Offerman (2003)’s model of a first price auction in which common and private value elements enter additively into bidders’ valuations. This framing is useful to capture the fact that in a drawdown, potential buyers differ both in their beliefs about future crude oil prices and in their opportunity costs of purchasing SPR crude. The latter generally differ from publicly reported spot prices because of transportation costs, long-term supply contracts, inventory levels, and other idiosyncratic factors. I use this basic model to show how price indexing can benefit the seller both by increasing revenue and by reducing the likelihood of default. I

then extend the model to address additional features common to markets for crude oil and other commodities, including the possibility of hedging outside the auction mechanism and the sale of multiple units via uniform price auctions.

Empirically, the model predicts that there should be no evidence of common values if the price index coincides with the common component of bidders' valuations (up to a shift of location). I test this prediction using two complementary strategies. First, I show that greater valuation uncertainty is not associated with greater bid shading, consistent with the absence of private information (Nyborg et al., 2002). To quantify uncertainty, I exploit the fact that the time elapsed between the auction and delivery varies considerably within the sample. Second, I show that there is no statistically significant within-auction correlation in bids after controlling for participation and other observables. This result is consistent with independent private values and hence the absence of private information (Hickman et al., 2021). Although both results are negative, they together provide suggestive evidence against strong adverse selection after price indexing. Furthermore, they suggest that DOE could not have significantly improved revenue by choosing a different price index. Because these results may reflect sampling variability, I provide numerical evidence that the second test has favorable power in multiunit settings even for modestly sized samples.

While the test results are consistent with the absence of common values after price indexing, they do not establish that common values would have been present had DOE adopted a different auction protocol. Thus, complementary evidence of private information is necessary to establish a revenue impact from price indexing. There are some reasons to think that firms bidding for SPR crude, primarily large domestic oil refiners, may be privately informed about (local) crude oil prices. While it is unlikely that individual firms hold much private information about oil prices in general, contract pricing for physical delivery can deviate significantly from benchmarks depending on time, location, and crude oil specifications. Pricing in the physical market is not fully transparent as parties generally have no reporting obligations (Fattouh, 2011). Moreover, the US oil refining sector has become relatively concentrated in recent decades. Taking this into account, I argue that individual buyers are plausibly informed about future supply and demand shocks relevant to pricing US-delivered physical contracts, such as their own future operational decisions.

Detecting the presence and significance of this type of private information is difficult for two reasons. First, the theoretical analysis suggests that the bidding data itself is not informative about bidders' underlying beliefs about oil prices. This interpretation is consistent with the empirical results above. Second, there is limited policy variation as DOE has used similar price adjustment mechanisms since the creation of the SPR in 1975. I therefore take an indirect approach and leverage the model to assess the minimum amount of private infor-

mation for which price indexing could have significantly increased revenue relative to a fixed price counterfactual. There is greater reason to believe that the impact of price indexing in 2022 was economically significant if this “bound” is small than if it is large.

The main challenge in constructing such a bound is the difficulty of characterizing equilibria in pay-as-bid auctions under realistic assumptions on bidder valuations and supply. To avoid this challenge, I instead analyze a hypothetical uniform price auction with price indexing. Building on Vives (2010), I show that the revenue benefit of price indexing in the uniform price setting depends on two sufficient statistics: the slope of bidders’ marginal valuation functions and the ratio of the precisions of bidders’ private information and the idiosyncratic cost distribution. I calibrate the slope parameter using bids from the 2022 drawdown under the assumption that revenue in pay-as-bid and uniform price auctions is similar on average (an approximation which I argue is reasonable based on well-known results from the multiunit auctions literature). The model then pins down the minimum ratio of precisions consistent with revenue impacts of a given size. I conclude that private information amounting to 5-15% of the total variation in bidder’s ex ante information would suffice to increase seller revenue by \$0.25-\$1.00 per barrel. In the context of the 2022 drawdown, such a gain corresponds to tens of millions of dollars in revenue.

The results of this paper are potentially relevant for sellers in other contexts and for recent debates surrounding the use of fixed prices in DOE’s crude oil purchases to fill the SPR.¹ The analysis suggests that price indexing can be useful when three conditions are met. First, buyers’ valuations must have an important time-varying common component (in this case, the comovement of opportunity costs as oil prices change over time). Second, the time between auction and delivery must be long enough that valuations might change significantly through this component. Third, it must be possible to construct a suitable price index. These conditions plausibly obtain in many commodity markets. As discussed below, price indexing is a common feature of many privately-negotiated long-term contracts, especially in the commodities sector. Perhaps surprisingly, there are few examples of formalized basis auctions in the empirical auctions literature.² Future research may be useful in further clarifying the tradeoffs associated with price indexing and the circumstances in which sellers might benefit (or not) from formal basis auctions.

¹DOE has historically used index-based price adjustment in crude oil procurements. In 2022, DOE unsuccessfully experimented with fixed price procurements. I discuss this policy change in Section 3.

²When public sector entities participate in commodities markets, fixed price formats are often used. Examples include sales of gold by the UK treasury (Wetherilt and Young, 2003), bulk grain tenders (Heigermoser et al., 2020), and electricity procurement (Cleary and Bishop Ratz, 2021).

Related literature Price indexing is distinguished from other forms of price adjustment by the use of pre-specified formulae to adjust final payments based on publicly available information (often spot market prices). This contrasts with cost-plus contracting, for example, where final payments are adjusted according to one party’s reported costs.³ Price adjustment mechanisms of various forms are ubiquitous in long-term contracting. In a classic series of empirical papers, Paul Joskow describes the extensive use of both price indexing and cost-plus clauses in electric utility coal procurement contracts (Joskow, 1985, 1988, 1990). Building on work by Goldberg (1985) and Goldberg and Erickson (1987), Joskow argues that price adjustment mitigates ex post opportunism, keeps moral hazard in check, and preserves incentives for cost-minimization over the life of a contract. Price indexing (also called formula pricing) is commonly found in contracts for physical delivery of crude oil and refined products (Fattouh, 2011), among other commodities such as natural gas and grain. The use of price indexing is not limited to commodities markets. In another classic paper, Crocker and Reynolds (1993) document the use of so-called “Economic Price Adjustment” clauses in US Air Force engine procurement. Acquatella et al. (2023) analyze price indexing in reimbursement for physician-administered drugs under Medicare Part B.

Price indexing specifically has received limited attention in the empirical auctions literature. In a notable exception, Kosmopoulou and Zhou (2014) find that the introduction of escalation clauses for bitumen (asphalt) costs in Oklahoma highway procurement contracts was associated with more aggressive bidding. Howell (2020) finds that similar clauses in Kansas highway procurement contracts reduced the sensitivity of bids to oil prices.⁴ It is not obvious why more work of this kind has not emerged given that auctions are often used to allocate long-term contracts, and long-term contracts often feature price indexing. Indeed, Kosmopoulou and Zhou (2014) report that forty states use some form of price adjustment for bitumen and fuel costs in highway procurement, a fact which has often been neglected in studies of highway procurement. Relative to these works, the main contribution of this paper is to formulate and empirically evaluate the prediction that price indexing reduces the significance of private information about common elements of bidders’ valuations.

Royalties and related forms of securitization constitute a different form of price adjustment (DeMarzo et al., 2005; Che and Kim, 2010). Kong et al. (2022) estimate a model of cash-royalty auctions in which bidders receive affiliated, bidimensional private signals. The information structure in this paper can be viewed as a simplification of their more general

³Bajari and Tadelis (2001) rationalize cost-plus contracting as an optimal response to uncertainty arising from project complexity (e.g., in construction). In contrast, price indexing tends to be used for routine or predictable transactions where market prices constitute the main source of uncertainty.

⁴Both papers emphasize that price adjustment shapes the allocation of risk between buyers and sellers. In contrast, I maintain the assumption of risk neutrality.

structure: bidders receive two signals, but “types” are one-dimensional and all information enters additively into bidders’ payoffs (Goeree and Offerman, 2003). Another key difference with Kong et al. (2022), and related work such as Bhattacharya et al. (2022), is the absence of moral hazard: setting aside the possibility of default, bidders’ ex post actions have no impact on surplus or bidding strategies, and hence do not inform auction design.

One factor that may have constrained prior empirical work on price indexing is the relative scarcity of data on bids for long-term contracts outside formal auctions in regulated settings. From this perspective, bidding data from the 2022 SPR drawdown might be viewed as a window into the broader market for long-term bilateral commodities contracts. The information contained in the data is potentially illuminating to the extent that SPR sale procedures and contracts resemble the informal negotiations and bilateral contracts by which oil refiners and other buyers maintain supply portfolios in the regular course of business.

The management of the SPR has become an important domain of public debate in recent years. This paper provides the first microeconomic analysis of the sales procedures used in the historic 2022 SPR drawdown. In doing so, this paper complements prior work analyzing the macroeconomic impacts of SPR releases (Newell and Prest, 2017; Kilian and Zhou, 2020; Stevens and Zhang, 2021). In the only prior retrospective analysis of the 2022 drawdown that I am aware of, Razek et al. (2023) argue that the 2022 release may have inadvertently contributed to gasoline price increases. Earlier work in operations research addresses optimal procurement and drawdown policies for petroleum reserves while abstracting from auction design (Teisberg, 1981; Oren and Wang, 1986).

Structure of the paper Section 2 presents the model and extensions. Section 3 introduces the data and describes the SPR drawdown procedure in detail. Section 4 presents the tests for common values. Section 5 presents the bounding exercise. Section 6 concludes. Proofs are contained in the Online Appendix.

2 Model

This section presents a generic model of auctions with price indexing highlighting potential benefits to the seller. I use the model to develop intuition for the empirical analysis of the 2022 SPR drawdown, which constitutes the remaining sections of the paper.

A seller is endowed with a single, indivisible good which cannot be immediately delivered to a buyer. In order to sell the good, the seller will hold an auction at time $t = 0$ for delivery at time $t = 1$. There are N symmetric, risk neutral potential buyers who differ in their opportunity costs of purchasing the good. Buyer i ’s valuation is the opportunity cost of

purchasing a substitute good at time $t = 1$. This opportunity cost is given by $w_1 + u_i$, where w_1 is the realization of a stochastic price index W_t at $t = 1$ and u_i is a *cost basis*.⁵

Each potential buyer i privately observes his own cost basis u_i . In addition, i receives a private signal ω_i of the true but not-yet-known ex post index price. I impose the following technical conditions on the distributions of u_i and ω_i , which allow potential buyers to be ordered according to a scalar “type” $s_i = \omega_i/N + u_i$ (Goeree and Offerman, 2003).

Assumption 1. (i) u_i is drawn independently across buyers from a distribution F_u having log-concave density f_u with support on a bounded interval $[\underline{u}, \bar{u}]$; (ii) ω_i is drawn independently across buyers from a distribution F_ω having log-concave density f_ω with support on a bounded interval $[\underline{\omega}, \bar{\omega}]$ and satisfying $E[\omega_i] = w_1$; (iii) ω_i is independent of u_i .

(ii) amounts to the assumption that buyers lack an informative prior for the distribution of W_1 before observing their private signals $\omega_1, \dots, \omega_n$. Because the private signals are distributed independently, the mean signal $\bar{\omega} = \frac{1}{N} \sum_{i \leq N} \omega_i$ is the most efficient estimator of w_1 that can be constructed from the signals $\omega_1, \dots, \omega_n$. In this sense $\bar{\omega}$ can be viewed as the common component of buyers’ valuations at the time of the auction.

The seller chooses an auction format. One natural candidate is a first price auction in fixed prices. Suppose there is no reserve price and that ex post default is impossible. Goeree and Offerman (2003) characterize a pure strategy Bayes Nash equilibrium of this game:

Lemma 1. Let $y_i = \max_{j \neq i} s_j$. The N -tuple of strategies $(\beta_1(\cdot), \dots, \beta_N(\cdot))$, where

$$\beta_i(x) = E[\bar{\omega} + u_i | s_i = x, y_i \leq x] - E[x - y_i | s_i = x, y_i \leq x], \quad (1)$$

is an equilibrium of the first price auction in fixed prices.

The strategy described by equation (1) has a standard interpretation. The first term represents i ’s interim assessment of his opportunity cost conditional on winning the auction. This term reflects a potential buyer’s anticipation of the winner’s curse: the auction winner learns that other bidders had lower types, and hence that $\bar{\omega}$ is less than ω_i . The second term represents i ’s information rent. This is the surplus buyer i earns on account of being privately informed about his own type (in this case the scalar $s_i = \omega_i/N + u_i$).

An alternative to a fixed price auction is a *basis auction*. In a basis auction, bids are expressed as a differential to a floating index price chosen by the seller. In a first price basis

⁵Throughout the paper I assume that u_i is independent of the realization of the index price W_t . Hence, buyer i ’s cost basis does not vary with the oil price level. This assumption is potentially restrictive. Below, I allow the index price chosen by the seller to differ from W_t . In this way, the model can accommodate additively separable common shocks to buyers’ costs bases that are correlated with changes in oil prices. However, the model cannot accommodate buyer-specific cost shocks driven by changing oil prices.

auction, the highest differential wins, and the winner's final payment is the sum of his or her bid and the realization of the index price upon delivery at $t = 1$. If W_t specifically is chosen, the first price basis auction reduces to a first price auction with independent private values (IPV). The pure strategy Bayes Nash equilibrium of such a game is well-known:

Lemma 2. *Let $y_i = \max_{j \neq i} u_j$. The N -tuple of strategies $(\delta_1(\cdot), \dots, \delta_N(\cdot))$, where*

$$\delta_i(x) = E[y_i | y_i \leq x], \quad (2)$$

is an equilibrium of the first price basis auction with index price W_t .

Unlike (1), the strategy (2) does not depend on buyers' private signals $\omega_1, \dots, \omega_N$. As a result, the seller's expected revenue improves, for two reasons. First, buyers' "types" (now simply the private costs $u_1, \dots, u_N$) are less dispersed, shrinking information rents. Second, there is no winner's curse (adverse selection). Moreover, total surplus increases because the private signals no longer affect the allocation of the good: the highest cost basis u_i wins.⁶

Proposition 1. *Expected revenue and total surplus are greater in the first price basis auction.*

I prove Proposition 1 using properties of the order statistics of samples from log-concave distributions. The intuition can be stated in terms of Milgrom and Weber (1982)'s linkage principle: price indexing tightens the statistical linkage between a buyer's information and his expected payment, increasing the seller's expected revenue.

Proposition 1 offers one important rationale for why a seller might prefer the basis auction to a fixed price auction. A second relates to opportunistic ex post behavior. In particular, the basis auction reduces buyers' incentive to default or seek renegotiation after low realizations of W_1 . After a fixed price auction, a winning bidder i is incentivized to default if $w_1 + u_i$ is lower than $\beta_i(s_i)$ upon delivery. After a basis auction, there is no such incentive: because $\delta_i(u_i) \leq u_i$, a winning bidder always benefits from taking delivery. The potential for price indexing to prevent opportunistic ex post behavior is a central emphasis of the long-term contracting literature (Goldberg, 1985; Joskow, 1988).

Imperfect price indices A winner in the basis auction obtains what Joskow (1988) describes as a "market price" contract. He writes: "The primary problem with a market price contract is defining an appropriate market price norm[.]" In other words, it may not be possible for the seller to index bids using the "ideal" index price W_t .

I now suppose the seller chooses a price index M_t which does not necessarily coincide with W_t . Buyers' private signals now reveal some information about the index M_t and some

⁶Efficiency need not hold in the case of buyer asymmetry (Maskin and Riley, 2000).

information about the residual common value $Z_t := W_t - M_t$. Supposing this information can be disaggregated, I now assume buyers observe distinct, independent private signals μ_i and ζ_i corresponding to the $t = 1$ realizations m_1 and z_1 , respectively.

Assumption 2. (i) μ_i is drawn independently across buyers from a distribution F_μ having log-concave density f_μ with support on a bounded interval $[\underline{\mu}, \bar{\mu}]$ and satisfying $E[\mu_i] = m_1$; (ii) ζ_i is drawn independently across buyers from a distribution F_ζ having log-concave density f_ζ with support on a bounded interval $[\underline{\zeta}, \bar{\zeta}]$ and satisfying $E[\zeta_i] = z_1$; (iii) ζ_i and μ_i are mutually independent and independent from u_i .

In this case, the equilibrium bidding strategy in the first price basis auction is:

$$\delta_i(x) = E[\bar{z} + u_i | \tilde{s}_i = x, y_i \leq x] - E[x - y_i | \tilde{s}_i = x, y_i \leq x] \quad (3)$$

where $\tilde{s}_i = \zeta_i/N + u_i$ and $y_i = \max_{j \neq i} \tilde{s}_j$. If the residual common value Z_t is uncertain, bidders earn information rents from ζ_i and the basis auction fails to eliminate the winner's curse. Furthermore, the buyer with the highest private cost basis u_i is no longer certain to win, reducing total surplus. Nevertheless, seller revenue is greater than in a fixed price auction so long as the distribution of the private information about the residual common value Z_t is less dispersed than the distribution of private information about W_t .

Hedging In general, the ability to hedge outside the auction mechanism could affect equilibrium bidding strategies and hence the comparison of auction formats. Here I present a brief argument that the existence of a futures market specifically will not necessarily affect the seller's comparison of auction formats above provided that buyers are risk neutral and futures markets are efficient. A similar argument can be applied to other hedging instruments, such as forwards.

Suppose bids are publicly disclosed immediately after the conclusion of the auction, revealing all buyers' private information. There is a futures contract for the delivery of one unit at time $t = 1$. Buying and selling the futures contract is costless for auction participants, including short-selling, subject to an exogenous limit on position size. Auction participants are price-takers in the futures market. Let the random variable F_1^s denote the price of the contract and Ω_1^s the set of public information at time s . The argument relies on two implications of no arbitrage. First, bidders' private information must be aggregated into the futures price immediately: letting $0+$ denote the moment after the auction concludes, we have $F_1^{0+} = E[W_1 | \Omega_1^0, \bar{\omega}]$. Second, the contract must settle at the index price: $F_1^1 = W_1$. Consequently, no winning or losing bidder can earn profits in expectation from buying or selling the contract after the auction concludes. The optimal bidding problem is unchanged

both for the fixed price and the basis auction formats, and equilibrium bidding strategies remain the same.

Two caveats should be made. First, the role of financial contracts could be significantly different if buyers were risk averse. Risk aversion provides a natural motivation for hedging, but can significantly change equilibrium bidding strategies in auction models. Second, the argument makes clear that ω_i should be interpreted as private information which is not already aggregated into futures prices via Ω_1^0 at the time of the auction.

Multiple units Price indexing is often used when transacting commodities in bulk. The uniform price and pay-as-bid auctions are two common generalizations of the first price auction to the case of multiple or divisible goods. The uniform price auction admits well-behaved symmetric linear equilibria for many Gaussian information structures. I therefore focus on the uniform price case, comparing the basis and fixed price formats.⁷

Suppose there are exactly Q identical (and divisible) goods for sale. Departing from the model above, I now assume that buyer i 's marginal opportunity cost takes the form:

$$v_i(q) = w_1 + u_i - \lambda q \quad (4)$$

where $\lambda > 0$ captures diminishing marginal benefits from larger purchases. The assumption of diminishing marginal benefits is standard in the multiunit auction literature and is often motivated by opportunity costs, inventory costs, or disutility from risk. The assumption that λ is the same for all buyers simplifies the equilibrium analysis below but is not essential.

For this section only, I parameterize the general log-concave distributions F_u and F_ω in Assumption 1 while dropping the bounded support assumptions:

Assumption 3. (i) u_i is drawn independently across buyers from a Normal distribution with mean θ_u and variance σ_u^2 ; (ii) ω_i is drawn independently across buyers from a Normal distribution with mean w_1 and variance σ_ω^2 ; (iii) ω_i is independent of u_i .

Models combining linear marginal valuations and Gaussian information structures are widely used in the theory literature on imperfectly competitive financial markets (Rostek and Yoon, 2023). The case of uniform price auctions with fixed supply is analyzed in Vives (2010) and Vives (2011). In Vives (2010) and Vives (2011) buyers observe one signal of a composite type encompassing both common and private value elements. The model presented here

⁷Equilibrium analysis of the pay-as-bid auction has proven challenging despite its wide usage in sovereign debt auctions and other settings (Hortacsu and McAdams, 2018; Pycia and Woodward, 2025). Wittwer (2018) and Allen and Wittwer (2025) present an interesting closed form model of a pay-as-bid auction with private information which could serve as a fruitful starting point for extending the analysis in this paper, at the cost of a stochastic supply assumption. (I thank a referee for this reference.)

differs by assuming that buyers' information includes distinct common and private value signals, requiring a slightly different equilibrium characterization.

The main goal of this section is to demonstrate that the revenue advantage of the basis auction in Proposition 1 persists in a multiunit environment. Part (i) of the following result shows that this is the case for the uniform price auction when N is sufficiently large. In addition, I characterize the difference in expected revenue, which will be used in Section 6. For completeness, part (ii) shows that the difference becomes smaller when buyers have less private information about W_t .

Proposition 2. (i) Fix $r = \sigma_u^2/\sigma_\omega^2 > 0$. For large N , the seller's expected revenue is greater in the uniform price basis auction than the uniform price auction in fixed prices. The difference in expected revenue per unit is:

$$\Delta = \lambda \left(\frac{Q}{N} \right) \Psi(r, N) \quad (5)$$

for a known $\Psi(r, N) > 0$ (see Online Appendix A.2). (ii) For fixed N , $\lim_{s \rightarrow \infty} \Psi(s, N) = 0$.

In a uniform price auction, buyers compete by submitting downward-sloping demand schedules. The optimal demand schedule maximizes a buyer's expected profits at each possible clearing price. In a uniform price basis auction indexed to W_t , the interim expected profit of a buyer i who wins q units when d is the basis price that clears the auction is

$$E[\pi_i | \omega_i, u_i, d] = (u_i - d)q - \frac{\lambda}{2}q^2$$

which shows that buyer i infers no useful information from the auction-clearing basis price. Hence, there is no adverse selection. In the unique symmetric linear equilibrium, buyer i submits a demand schedule

$$X(d, \omega_i, u_i) = \left(\frac{1}{\lambda + \psi_{basis}} \right) \cdot (u_i - d) \quad (6)$$

where $\psi_{basis} = \frac{1}{N-1} \left(\frac{N-1}{N-2} \right) \lambda > 0$ is referred to as price impact.

In contrast, the uniform price auction in fixed prices does not eliminate adverse selection, resulting in less aggressive bidding. In this case, interim expected profit is

$$E[\pi_i | \omega_i, u_i, p] = (E[\bar{\omega} | \omega_i, u_i, p] + u_i - p)q - \frac{\lambda}{2}q^2$$

Assumption 3 implies that $E[\bar{\omega} | \omega_i, u_i, p]$ is linear in the conditioning variables. Under the hypotheses of Proposition 2, there is a symmetric linear equilibrium in which each buyer i

submits a demand function

$$X(p, \omega_i, u_i) = \left(\frac{1}{\lambda + \psi_{fixed}} \right) \cdot \{E[\bar{\omega}|\omega_i, u_i, p] + u_i - p\} \quad (7)$$

with price impact ψ_{fixed} . A comparison of expected revenue per unit given the equilibrium strategies (6) and (7) gives (5). (See Online Appendix A.2 for details.)

Proposition 2 does not directly speak to whether price indexing would improve the seller’s expected revenue in a pay-as-bid environment. However, a significant body of work in the empirical auctions literature points to what Pycia and Woodward (2025) describe as a “rough revenue equivalence” between the pay-as-bid and uniform price formats – at least in the case of fixed price private values auctions, seller revenue is often similar across formats.⁸ If one expects the pay-as-bid basis and fixed price auctions to generate similar revenue as their uniform price counterparts, Proposition 2 can be taken to suggest that price indexing is likely to improve seller revenue in the pay-as-bid case as well.

2.1 Connection to SPR drawdowns

Sales from the SPR can be viewed as a pay-as-bid basis auctions with potentially imperfect price indices. The model and extensions above suggest that the bidding data ought to exhibit little evidence of adverse selection unless buyers hold significant private information about the residual common value Z_t not captured by the price index M_t . This section briefly discusses potential sources of private information in SPR drawdowns.

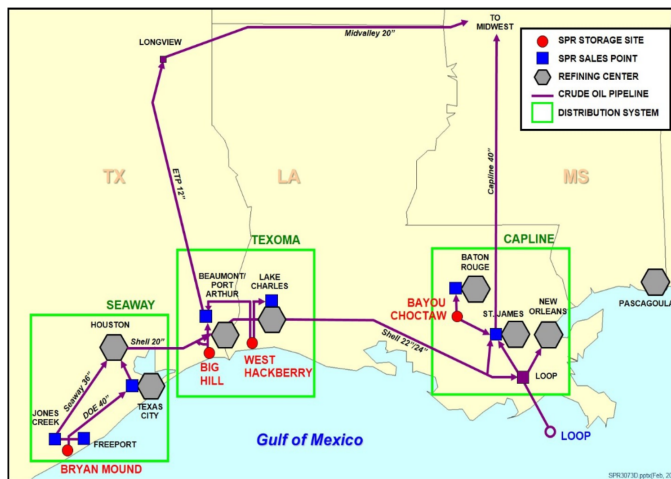
The SPR was created by the 1975 Energy Policy and Conservation Act (ECPA) in response to the 1973 oil crisis. Currently, the SPR stores up to 713.5 million barrels (MMbbl) of crude oil at four distinct locations along the US Gulf Coast, shown in Figure 1. Each location stores a combination of sweet (i.e., low sulfur) and sour (i.e., high sulfur) crude oils in underground salt caverns. Locations differ in their capacities, crude oil characteristics, access to pipelines and shipping terminals, and proximity to major Gulf Coast refineries.

During an SPR drawdown, DOE uses pay-as-bid basis auctions to sell millions of barrels of oil at once. In recent sales, two distinct price indices have been used depending on the type of oil. For sweet crude, the price index is the average over the five days surrounding delivery of the Argus WTI Houston price; for sour crude, Argus Mars is used instead.⁹

⁸Marszalec (2017) Table 10 compiles a summary of point estimates from the literature. Revenue differences of less than 5% have been found in most prior studies, albeit typically in the sovereign debt context where the number of bidders is often large. Ausubel et al. (2014) prove that there is no general revenue ranking.

⁹WTI Houston Weighted Average Month 1, Houston Close and Mars Weighted Average Month 1, Houston Close. “Weighted Average Month 1” refers to the average reported transaction price for front month physical contracts.

Figure 1: SPR System Map



Source: DOE. Seaway 36" refers to the Seaway Jones Creek-ECHO lateral.

The bases between Argus WTI Houston, Argus Mars, and WTI Cushing (a key benchmark in futures markets) can be significant. During 2022, the basis between WTI Houston and WTI Cushing ranged from \$0.54 to \$2.50 per barrel, while the basis between Mars and WTI Cushing ranged from $-\$6.16$ to $\$0.26$ per barrel.¹⁰ These differences primarily reflect regional heterogeneity in supply and demand for different types of crude oil. Issues related to measurement can also play an important role. Although Argus WTI Houston and Argus Mars are widely accepted benchmarks for sweet and sour crude on the US Gulf Coast, it is important to understand that these (and other) oil price benchmarks represent “assessed” prices. Buyers and sellers of physical crude are not required to report transaction prices. Firms often choose to report transaction prices voluntarily, but the indices published by commercial price reporting agencies (PRAs) can also incorporate non-price information such as bids and offers or even interviews with market participants. Assessed prices are potentially vulnerable to manipulation and may not accurately reflect market conditions if there is reporting bias or if the PRA’s modeling assumptions are incorrect (Fattouh, 2011).

Buyers in this environment may be privately informed both about the price index M_t and the residual common value Z_t . Buyers are typically large oil refiners or commodities marketing firms. Within regional markets for physical crude, large buyers and sellers may be able to anticipate the effects of their own strategic decisions (for example, refinery utilization) on M_t and Z_t . Many oil refiners are vertically integrated into marketing, storage, or production, each of which could generate additional information. Beyond this, buyers

¹⁰Closing prices for WTI Houston (Argus) vs. WTI Trade Month Futures (Bloomberg: HRT1 Comdty) and Mars (Argus) vs WTI Spread Trade Month Swap (Bloomberg: YVA1 Comdty).

may be incentivized to strategically withhold transaction prices or other information from the PRA.

Two potentially important components of the residual common value Z_t are transportation costs and crude oil quality. In order to take delivery of SPR crude, a buyer must typically secure capacity on one or more commercial pipelines.¹¹ Most pipeline capacity is allocated via firm, long-term contracts. Winning bidders that do not already hold firm capacity on pipelines connecting to SPR facilities must quickly purchase capacity in secondary markets, which are prone to scarcity pricing. Buyers may differ in their beliefs about future prices in these markets, which can shift the value of SPR crude for many firms. Buyers may also differ in their beliefs about the quality of SPR crude. Buyers who have previously purchased SPR crude may have more information about the physical characteristics of crude stored at particular SPR sites. As discussed below, DOE attempts to mitigate quality concerns through additional price adjustments, but these price adjustments are not tied to market prices and may not accurately reflect buyers' valuations for crude oil characteristics such as sulfur content and API gravity.

In addition to any effects on Z_t , transportation costs are likely the predominant source of variation in the cost basis u_i . SPR facilities are more easily accessed by some potential buyers than others. As discussed above, obtaining pipeline capacity in secondary markets can be costly for buyers that do not already hold firm capacity reservations. Maritime shipment via Jones Act vessels is subject to similar dynamics. Other potentially important sources of variation in u_i include buyers' inventories and non-SPR supply contracts. A buyer that holds lower-cost supply contracts benefits less from purchasing SPR crude at the margin. For an oil refiner, the value of SPR crude can also differ depending on output mix (gasoline vs. other distillates) and the demand for those outputs as well as the supply of complementary inputs.¹² In the multiunit context, a positive λ value is reasonable if the opportunity cost at which a buyer obtains SPR crude declines with purchase quantity. For example, shipping costs may increase non-linearly with quantity, reducing the benefit of larger purchases.

The analysis above assumes that a buyer's cost basis u_i is fixed across formats. If hedging is costly, this assumption can be violated in a manner that favors the basis auction. Oil refiners often attempt manage forward price risk by hedging the differential between crude oil and refined product prices (the "crack spread"). Refined products are typically sold in bulk on a floating price basis (for example, as a differential to WTI Cushing). In this situation, winning bidders in the basis auction are naturally hedged without the need to

¹¹Indeed, the necessity of allowing winning bidders time to arrange shipment is one important reason that delivery windows do not commence immediately.

¹²SPR crude would typically be blended with crude from other sources before distillation. Refineries are engineered to process unique blends ("slates") of crude oils having specific chemical properties.

obtain additional financial contracts, increasing the value of SPR crude.¹³

3 The 2022 Emergency Drawdown

In 2022, United States initiated seven large sales (or “drawdowns”) from the SPR. This section provides further background on SPR sales and introduces the data.

Historical background The main purpose of the SPR is to “protect the U.S. economy from severe petroleum supply interruptions” by distributing stockpiled oil in times of crisis (DOE, 2024). Emergency sales are triggered by presidential declaration. Prior to 2015, emergency sales were conducted infrequently and had never been larger than about 30 million barrels (MMbbl).¹⁴ The 2022 decision to sell 180 Mbbl (roughly one third of SPR inventory) in response to the Russia-Ukraine conflict was therefore a notable departure from prior policy.

To contextualize oil market conditions in 2022, Figure 3 plots the time series of daily WTI Cushing prices (panel a) and 30-day realized volatility (panel b) from 2000 to present, both in 2024 dollars. The time period spanning the 2022 drawdown is shaded in gray. WTI Cushing prices increased rapidly early in 2022, with prices reaching \$137 per barrel (2024 dollars) and 30-day realized volatility peaking at 82% on an annualized basis.

3.1 The mechanics of SPR sales

DOE has used its current sales procedures with only minor modification since 1983.¹⁵ Each sale begins with the issuance of a Notice of Sale. In the Notice of Sale, DOE stipulates target volumes of sweet and sour crude oil to be sold from designated storage sites, as well as crude specifications, logistical details concerning delivery, and other contract terms.

Figure 2 provides an example sales specification taken from the third sale in 2022. In this sale, which occurred on June 1, DOE sought to release both sweet and sour crude from the Bayou Choctaw storage site near Baton Rouge, LA. For the sweet crude “master line item” (MLI), 1,100 thousand barrels (Mbbl) were offered for delivery between June 21 and June 30. For the sour crude MLI, 6,000 Mbbl of sour crude were offered for delivery between July 1 and August 15. The rate at which crude oil can be transferred from storage to particular “points of sale” is limited by physical transfer capacity and scheduling constraints. The “delivery line

¹³Although crude oil futures markets are among the deepest financial markets in existence, taking futures positions is not costless due to margin requirements, maturity mismatch, and other factors.

¹⁴Notable “emergency” releases occurred after Operation Desert Storm (17.3 MMbbl), Hurricane Katrina (11 MMbbl), and the 2011 Libyan crisis (30.6 MMbbl).

¹⁵<https://www.spr.doe.gov/reports/SSPs/PART625.html>

Figure 2: Notice of Sale Example

Crude Oil Stream MLI	DLI - Mode of Delivery	Delivery Period	MLI Qty (MB)	MBD Avg	DLI Qty (MB)	MIN Qty (MB)
Bayou Choctaw Sweet 007		6/21-6/30	1,100			
	DLI-A Pipeline St. James Pipeline	6/21-6/30		300	1,100	350
	DLI-H Pipeline Baton Rouge (Bourre)	6/21-6/30		200	225	225

Crude Oil Stream MLI	DLI - Mode of Delivery	Delivery Period	MLI Qty (MB)	MBD Avg	DLI Qty (MB)	MIN Qty (MB)
Bayou Choctaw Sour 008		7/1-8/15	6,000			
	DLI-A Pipeline St. James Pipeline	7/1-8/15		300	4,300	350
	DLI-B Vessel Saint James Dock	7/1-8/15		200	800	250
	DLI-H Pipeline Baton Rouge (Bourre)	7/1-8/15		200	900	225

Source: DOE Notice of Sale (Emergency Drawdown No. 2B, May 24, 2022).

items” (DLIs) in Figure 2 indicate that pipeline delivery of sweet crude to the Baton Rouge (Bourre) point of sale was limited to 225 Mbbl of the total available 1,100 Mbbl. There was no binding limitation on delivery to the St. James point of sale. For sour crude, the same two points of sale were made available for 900 Mbbl and 4,300 Mbbl, respectively, while 800 Mbbl was only made available for vessel delivery (barges and tankers).

A bid specifies a quantity demanded and price for delivery of oil at a particular point of sale. Firms are permitted to submit multiple bids (i.e., a demand schedule). However, because of the logistics inherent to the physical transfer of oil, the minimum allowed bid size (indicated by the “MIN Qty” column in Figure 2) is relatively large.

For each MLI (crude oil stream), the auction is cleared as a pay-as-bid auction subject to DLI (point of sale) constraints: bids are fulfilled from high price to low price until the total MLI quantity is exhausted, unless a DLI constraint becomes binding, in which case any remaining MLI quantity is allocated to unconstrained DLIs.¹⁶

As discussed above, an important feature of the SPR auction is that bids are not expressed in fixed prices, but rather as a basis to an index price specified by DOE. The window for taking delivery of purchased crude typically does not begin until a few weeks after the auction, and can last more than a month (as in the Bayou Choctaw example above). During

¹⁶In the previous example, if the high bidder for Bayou Choctaw sweet requested 225 Mbbl at Baton Rouge (Bourre), leaving no remaining supply, then any further bids for delivery at Baton Rouge (Bourre) would be discarded and the remaining 875 Mbbl could only be sold to bidders requesting delivery at St. James. On the other hand, if the high bidder requested 1,100 Mbbl at St. James, no further supply would be available for bidders requesting delivery at Baton Rouge (Bourre). Similar dynamics arise in competitive wholesale electricity markets due to transmission congestion.

this time, the index price can move considerably. To illustrate, Table 1 shows the average values of WTI Houston and Mars crude prices during delivery periods in comparison to the prevailing index price on the date the Notice of Sale was issued (the “base reference price” or BRP).¹⁷ After the March 8 sale, sweet and sour crude prices were 12% higher than the BRP on the average day in the delivery window. After the June 28 sale, both were more than 25% lower on the average day.

Reserve prices DOE reserves the right to reject bids that are “below 95 percent of the sales price, as estimated by the Government, of comparable crude oil being sold in the same area at the same time.” The 95% threshold can be interpreted as a nominal reserve price. I describe this reserve price as *nominal* because bids below it may still be accepted “if the Contracting Officer determines such action is necessary to achieve SPR crude oil supply objectives and such offered prices are reasonable.” The model presented in Section 2 does not incorporate reserve prices. However, as explained below, it appears that the nominal reserve price was not binding during the 2022 SPR drawdown.

Price adjustments for quality SPR crude is segregated into eight crude oil streams (sweet and sour at each location) spread across sixty salt caverns. Each cavern contains a mixture of crude oils accumulated from different sources over time. As the caverns are large, tall, and narrow, the crude oil within a single cavern eventually becomes well-mixed due to temperature gradients (Department of Energy, 2024). Nevertheless, the physical characteristics of crude oil delivered to a buyer at a specific place and time may differ from specifications disclosed in the Notice of Sale depending on the state of the specific caverns used. When this occurs, DOE adjusts final payments according to tables of pre-specified price differentials that scale with each 0.1 degree difference in API gravity and 0.01 percentage point difference in sulfur content from announced levels. In the framework of the model, these adjustments can be viewed as partially “indexing” for one potentially important component of the residual common value Z_t . However, price increments are fixed in advance rather than being tied to market benchmarks. There are no adjustments for discrepancies in other crude oil characteristics.¹⁸

Use of fixed prices in acquisitions DOE historically used price indexing for crude oil purchases to fill the SPR. In late 2022, DOE modified its acquisition procedures to permit

¹⁷The reported crude oil indices correspond to the Bloomberg tickers USCRMEHC (Crude Oil WTI Houston) and USCRMARS (Crude Oil Mars). It is reasonable to expect that these series are highly correlated with the specific commercial price indices used by DOE.

¹⁸This is not to say that other crude oil characteristics cannot affect buyer valuations. Some potentially important characteristics include acidity, viscosity, and metal content.

(but not obligate) fixed price offers. After the change of administration in 2025, DOE again modified its acquisition procedures, this time in order to prohibit fixed price offers.¹⁹

To date, DOE has purchased approximately 58 MMbbl since the 2022 drawdown concluded. The first post-drawdown Request for Proposal (RFP) on December 16, 2022 required participants to submit fixed price offers. This RFP proved unsuccessful: no awards were made. For all 26 subsequent RFPs, DOE indexed payments to market prices on the award date (or in some cases the three days subsequent to the award date). As awards are typically made within two weeks of offer submission, but weeks or months before delivery, this can be viewed as a relatively rigid form of price-indexing.

DOE’s post-drawdown purchases have been criticized by some. Out of 27 RFPs, 16 resulted in undersubscription or cancelation, with 7 outright failures (including the initial fixed price RFP). In light of this paper’s analysis, it is plausible that a more flexible indexing procedure could have attracted a greater number of offers, perhaps at lower prices. Yet the key factor explaining undersubscription was likely DOE’s aggressive price targets (initially presented as guidance, later formalized with a reserve price). A limitation of the present paper is that I do not model reserve prices.

3.2 Bidding data

This section briefly describes the bidding data (Department of Energy, 2026). To begin, Table 2 presents summary statistics on participation, demand, and winning bids for each crude oil stream (MLI) and delivery location (DLI). Sweet crude accounted for 45% of sales by volume. On average, 10.9 bidders submitted 3.7 bids each for a given sweet crude MLI, often submitting bids on multiple DLIs at once. Each bid is associated with a particular DLI and quantity demanded. Like in other multiunit auction settings, the multiple bids submitted by an individual bidder together constitute a downward-sloping demand schedule. The basis component of a bid is referred to as the price adjustment factor (PAF). On average the highest winning PAF for sweet crude was \$1.19 and the lowest was $-\$2.89$. In comparison, the average nominal reserve price (expressed as a PAF) was $-\$5.13$. Prices were generally higher for pipeline delivery in comparison to vessel delivery. Sour crude MLIs had only 8.0 bidders on average, with winning PAFs ranging from $-\$1.44$ to $\$1.85$ on average.

Figure 4 presents firm-level bid curves as well as the total demand curve among bids

¹⁹The stated rationale for the 2022 rule change was to “increase flexibility” in “circumstances in which a fixed-price acquisition would better meet [...] statutory objectives.” A press release from the White House on October 18, 2022 explained that, “This repurchase approach will protect taxpayers and help create certainty around future demand for crude oil. That will encourage firms to invest in production right now[.]” The justification for the 2025 rescission was that changes to permit the use of fixed-price contracts had “only served to unnecessarily create confusion in industry, which uses index-price contracts[.]”

submitted for one specific sweet crude DLI in the August 8 sale. In this case DLI constraints precluded demand spillovers across points of sale, simplifying the visual presentation. The figure illustrates that there is significant variation across bidders both in terms of maximum expressed willingness-to-pay (the bid intercepts) as well as in terms of bid slope. No single bidder demanded more than half of the available supply (indicated by the vertical red line), while total demand at any price exceeded supply by a factor of three.

Domestic oil refiners were by far the largest purchasers of crude oil in the 2022 SPR sales. 74% of the crude oil sold in the 2022 SPR releases was purchased by one of the four largest US oil refiners (Marathon, Valero, ExxonMobil, and Phillips 66) or by Motiva (a Saudi Aramco subsidiary which owns a large refinery in Texas). Most of the remaining crude oil was purchased by commodities marketers. In practice the distinction between refiner and marketer is not always sharp: some refiners operate trading desks, while some marketers buy and sell on behalf of refiners.

More than 13% of bids were submitted below the nominal reserve price (i.e., had a PAF less than -5% of the BRP), indicating that bidders may not have expected the reserve to be enforced. Consistent with this belief, 1.3% of winning bids had a PAF less than -5% of the BRP. However, the vast majority of bids below this threshold were unsuccessful.

Exclusive tranches Empirical methods for auctions typically assume that the unit of observation is a single, independent auction with a fixed number of (potential) bidders. The latter assumption can be violated in SPR sales due to the DLI constraints: with respect to a single DLI, competition may change discontinuously with quantity. For sweet crude in the example in Figure 2, St. James DLI bidders faced more competition for quantities above than below 875 Mbbl because quantities in excess of 875 Mbbl were also contested by Baton Rouge (Bourre) DLI bidders.

To abstract from complications arising from the DLI constraints, the primary unit of observation for the empirical analysis below is an *exclusive tranche*. An exclusive tranche is defined as a quantity of crude oil accessible only to bidders on a particular DLI. In the example, there is one 875 Mbbl exclusive tranche associated with St. James, but no exclusive tranche associated with Baton Rouge (Bourre). For sour crude in the example in Figure 2, there are three exclusive tranches (one for each DLI). In total the data contains 54 exclusive tranches (2.16 per MLI per sale). The bids in an exclusive tranche can be viewed as arising from a distinct pay-as-bid auction with a fixed number of bidders. Restricting attention to exclusive tranches involves little loss of data: 92% of the total quantity sold can be attributed to an exclusive tranche.

4 Tests of common and private values

In this section, I test a key prediction from the theoretical analysis: if the price index used to determine final payments is sufficiently similar to the common component of valuations, there should be no evidence of adverse selection. Such a result would be consistent with, but would not imply, the conclusion that price indexing generated material benefits for DOE during the 2022 drawdown. (I defer further discussion of this issue to Section 5.) On the other hand, positive evidence of adverse selection would suggest that DOE’s choice of index could potentially be improved.

4.1 Test of common values

Few formal tests of common values have been proposed for the pay-as-bid auction.²⁰ One exception is a reduced form test due to Nyborg et al. (2002).²¹ This test is based on the prediction that bidders in a pay-as-bid auction will tend to reduce the level and steepen the slope of their bids as common value uncertainty becomes more severe (Ausubel, 2004). Like in the first price auction, a winning bidder learns that other bidders received lower signals of the common value. In addition, a bidder who wins a larger number of units learns that other bidders received relatively worse signals (the “champion’s plague”). A negative correlation between bid levels and uncertainty, or a positive correlation between bid slope and uncertainty, is therefore consistent with common values (Nyborg et al., 2002).

I implement this test using two distinct measures of uncertainty. The first is the CBOE Crude Oil Volatility Index (or OVX). OVX is a measure of implied volatility constructed from the price of options on the USO ETF, which tracks WTI Cushing prices.²² WTI Cushing prices are highly correlated with, but not identical to, the index prices used by DOE. Hence, OVX should correlate with uncertainty about the index price M_t . Because price indexing reduces the value of information about M_t , one would not expect to find a strong correlation between the OVX and observed bid slopes or levels unless uncertainty about WTI Cushing is correlated with uncertainty about the basis between M_t and WTI

²⁰Hortaçsu and Kastl (2012) exploit the availability of an instrument shifting some bidders’ information in the context of Canadian treasury auctions. Bonaldi et al. (2024) estimate a parametric model of a uniform price auction in which bidders have both private and common values, enabling a decomposition exercise. I lack the data to implement the test of Hortaçsu and Kastl (2012), while the lack of convenient and flexible parametric models for pay-as-bid auctions precludes an approach similar to Bonaldi et al. (2024).

²¹Nyborg et al. (2002) study pay-as-bid sovereign debt auctions. Prior applications of the test include Bjønnes (2001), Elsinger and Zulehner (2007) and Mariño and Marszalec (2023). A similar approach was considered in Hortaçsu (2002).

²²Source: CBOE (2026). The OVX is similar to the better-known CBOE VIX, which approximates stock market volatility based on S&P 500 index options. See Chen et al. (2018) for further discussion of the OVX.

Cushing or with uncertainty about time-varying elements of the residual common value Z_t . The latter could occur if, for example, pipeline shipping costs were correlated with oil prices.

The second measure of uncertainty is defined as the number of days elapsed between the sale date and the midpoint of the DLI delivery window indicated in the Notice of Sale (see Figure 2 for an example). If uncertainty increases over longer time horizons, as one might expect, time-to-delivery correlates both with index price uncertainty and with uncertainty about time-varying elements of the residual common value Z_t . The test can detect the latter. For example, a longer time-to-delivery plausibly correlates with greater uncertainty about the crack spread (the difference between refined product and crude oil prices), which lacks a simple relationship with the oil price level but which affects refiner profits. On the other hand, time-to-delivery would not correlate with time-invariant elements of Z_t such as those stemming from differences in crude oil characteristics not addressed by DOE’s quality adjustments, unless the common value of those characteristics also changed over time.

Results Table 3 summarizes regressions of bid levels and slopes on both measures of uncertainty. Bid levels are measured using a firm’s quantity-weighted average PAF. As in Nyborg et al. (2002), the steepness of bid slopes is measured by the quantity-weighted standard deviation in PAFs across a firm’s bids. This provides a simple but robust non-parametric summary of bid slope. Two specifications for the bid level are included: one using bidder-tranche level average bids, and one using the tranche-level average of bidders’ within-tranche average bids. The latter is intended to address bias that may arise from correlation in the error terms within an auction. I also control for sulfur content, delivery method, tranche size, and SPR site.²³ Observations are weighted by the measure of uncertainty.

To complement the regression results, Figure 5 depicts the correlations between bid levels, bid slopes and each measure of uncertainty. I present two versions in each panel: the raw correlation (left) and that obtained after residualizing on the controls included in the main regressions (right).

In Table 3, the coefficients on uncertainty in the bid level regressions in columns (1), (2), (4), and (5) are all positive. Thus, bid levels appear to increase on average when uncertainty is greater. The point estimate in (4) implies that shifting the delivery window back by a week increases average bids by about \$0.15 per barrel. The point estimate in (1) indicates that an increase of similar magnitude is implied by a 5 point increase in OVX (corresponding to a 5 percentage point increase in annualized implied volatility). These results can be viewed as inconsistent with the presence of residual private information, although the effects

²³Together, these characteristics control for potential entry. Bjonnes (2001) argues that controlling for participation is important for distinguishing between bidding behavior driven by common value uncertainty and bidding behavior driven by risk aversion.

are small in magnitude and none is statistically significant. Correspondingly, there is little visual indication of strong correlation in panels (a) or (c) of Figure 5. The coefficients on uncertainty in the slope regressions in columns (3) and (6) are statistically significant, but differ in sign. A steeper slope corresponds to larger standard deviation in bid prices. Hence, the estimate in (3) implies that bids become steeper when the OVX is greater, while the estimate in (6) implies that bids become flatter when time-to-delivery is longer. The first result is consistent with the presence of private information; the second is not. Taken together, the results in Table 3 do not provide clear evidence of adverse selection.

4.2 Test of private values

In this section I consider a direct test of independent private values (IPV) due to Hickman et al. (2021). The central idea underlying this test is that, under the null hypothesis of IPV with no unobserved heterogeneity, there should be no residual correlation in bids after controlling for the number of bidders N and other observables.

In a single-unit context, this idea can be implemented by regressing observed bids on the mean of within-auction rival bids, observables, and a flexible function of N . IPV is rejected if the coefficient on within-auction rival bids is different from zero. This idea can be extended to the context of a multiunit auction by restricting attention to quantity levels that are common to all submitted bids. For example, if all bidders are willing to accept a single unit of the good, one might use the highest price bid submitted by each bidder (the “bid intercept”). I adopt the bid intercept approach here.

Hickman et al. (2021) report that the single-unit test has favorable power even in small samples. It rejects not only pure common values, but also various alternative models such as the affiliated private values (APV) model and IPV models with auction-level unobserved heterogeneity. In Online Appendix B, I verify the finite-sample power of the multiunit bid intercept test using simulated bidding data from a calibration of Vives (2011). The test rejects conditional independent private values (CIPV) and pure common values at high rates in sample sizes comparable to that of the 2022 bidding data.

DOE allows bidders to stipulate a minimum bid quantity. When a bidder’s highest price bid has a non-trivial minimum quantity, it is not clear that the associated bid price should be treated like an intercept: the bidder is not willing to accept a small number of units at the indicated price. To address this issue, I implement the bid intercept test separately for all exclusive tranches and for the subsample of exclusive tranches in which no such bids were submitted. In the former sample, I retain bids with high minimum quantities when calculating N but exclude them from the estimation sample and from the calculation of the

mean of rivals’ bids. Both regression samples further exclude tranches with fewer than two bidders, since in this case the mean of rivals’ bids cannot be calculated.

Table 4 reports the results. The main specification is reported in column (3). As in Richert (2024), this model includes a cubic polynomial of N . The coefficient on the mean rival bid is negative and not significantly different from zero at conventional levels. Thus, the test fails to reject the null of IPV. A similar result is shown column (5), which retains the tranches excluded due to bids with high minimum quantities in the manner described above. The remaining specifications test the robustness of the result in column (3). Column (1) omits all covariates. Column (2) includes all covariates, but omits the polynomial in N . In both cases, I obtain positive but insignificant coefficients. For the main results in column (3), standard errors are clustered at the tranche level. Column (4) reports standard errors calculated without clustering. This change does not cause the main estimate to become statistically significant. Together with the results above, these findings provide suggestive evidence against adverse selection.

5 Gains from price adjustment

The test results in the previous section do not reveal evidence of strong adverse selection in the 2022 bidding data. While consistent with the implications of the model, the absence of evidence of adverse selection does not imply that adverse selection was eliminated by price indexing. To evaluate the impact of price indexing in the 2022 drawdown, the key question is whether buyers’ are privately informed about the index price M_t . Some potential sources of private information about M_t are discussed in Section 2.1. However, because the model suggests that bidding strategies do not depend on this type of information, the bidding data itself cannot resolve this question.

In this section I address the nature of buyers’ information indirectly by attempting to quantify the amount of private information for which DOE’s revenue gains from price indexing would have been large during the 2022 drawdown. If only a small amount of private information would have been needed to generate large revenue gains, there is greater reason to believe that the impact of price indexing was economically significant. Such an exercise is inherently counterfactual and hence requires a model for which equilibrium analysis is tractable. I therefore adopt the uniform price framework from Section 2. This model can be calibrated to the bidding data. Although the use of a uniform price model entails a loss of realism, Proposition 2 offers a convenient closed form prediction of counterfactual revenue.

I proceed in two steps. First, I exploit the empirical regularity that uniform price and pay-as-bid auctions tend to generate similar seller revenue to calibrate the slope parameter

λ in bidders' valuations (4). Second, I use Proposition 2 to calculate the minimum amount of private information necessary for price indexing to generate a given increase in expected revenue in a set of hypothetical auctions comparable to the exclusive tranches in the bidding data. I find that relatively little private information would be necessary for the seller to obtain economically meaningful revenue gains from price indexing.

Calibration of slope λ The observed bidding data is generated by DOE's pay-as-bid basis auction. Consider a uniform price basis auction in which each bidder has the same λ parameter.²⁴ It can be shown that the expected auction-clearing basis price is

$$E[d] = \theta_u - \lambda \cdot \left(\frac{Q}{N} \right) \cdot \left(\frac{N-2}{N-1} \right).$$

If the expected revenue from a pay-as-bid basis auction is similar to the expected revenue from a uniform price basis auction, then the quantity-weighted average winning basis d_τ observed in exclusive tranche τ can be predicted by the linear model

$$d_\tau = x'_\tau \beta_{pred} - \lambda_{pred} \cdot \left(\frac{Q_\tau}{N_\tau} \right) \cdot \left(\frac{N_\tau - 2}{N_\tau - 1} \right) + u_\tau$$

where x_τ is a vector of tranche characteristics. Using linear regression weighting tranches by quantity, I obtain a point estimate $\hat{\lambda}_{pred} = 0.0022$ for λ_{pred} with 95% bootstrap confidence interval (0.0005, 0.0051).²⁵ In (4), such an estimate implies that bidders' marginal valuations decline by \$1.10 per barrel for an additional 500 Mbbl (the average size of observed awards).

Information structure analysis In the uniform price environment, the per-barrel expected revenue benefit from price indexing is given by (5) for a given slope parameter λ and precision ratio $r = \sigma_u^2 / \sigma_\omega^2$. These two parameters are sufficient to determine the tranche-level revenue impact conditional on Q_τ and N_τ . Figure 6 plots the ratio $\frac{1}{1+r} = \sigma_\omega^2 / (\sigma_u^2 + \sigma_\omega^2)$ versus the log of the quantity-weighted mean per-barrel revenue impact across all tranches. The ratio $\frac{1}{1+r}$ is interpreted as the fraction of variation in buyers' information $\omega_i + u_i$ attributable to private information about the common value W_1 . For a sense of scale, one would expect that the standard deviation of private costs σ_u is unlikely to exceed a few dollars per barrel. In the figure, the blue vertical line at zero corresponds to an expected gain from price indexing equal to \$1.00 per barrel. This occurs for $\frac{1}{1+r} = 0.15$ (95% bootstrap CI: (0.07, 0.38)),

²⁴Although generated in a pay-as-bid environment, the demand curves in Figure 4 are suggestive that λ may differ across bidders. I exclude this possibility in the interest of tractability and statistical efficiency.

²⁵For the regression x_τ includes Notice of Sale, location, sulfur content, and delivery mode (pipeline vs. vessel) fixed effects.

when σ_w is less than half as large as σ_u . For larger amounts of private information, the expected gain is greater.²⁶ The expected gain remains economically meaningful even for substantially smaller amounts of private information. The green vertical line corresponds to an expected gain from price indexing equal to \$0.25 per barrel. This occurs for $\frac{1}{1+r} = 0.04$ (95% bootstrap CI: (0.02, 0.25)), when σ_u is roughly five times greater than σ_w .

Over 180 million barrels, savings of \$1.00 per barrel (\$0.25 per barrel) sum to \$180M (\$45M). These results are suggestive that price indexing could have generated large revenue gains in the 2022 SPR drawdown even for small amounts of private information.

Market volatility If potential buyers’ private information about oil prices becomes more dispersed during times of elevated uncertainty, one would expect adverse selection in the fixed price auction to worsen, making the basis auction more attractive. If this is the case, the benefit from using a basis auction may be greater during emergency sales like those in 2022 than during non-emergency sales (such as DOE’s periodic budgetary sales).

6 Conclusion

This paper analyzes index-based price adjustment during DOE’s historic 2022 SPR drawdown. Drawing on the auctions literature, I use an illustrative model to show how price indexing can reduce information rents, limit adverse selection, improve efficiency, and reduce the risk of default. In the extreme case that the common component of bidders’ valuations coincides with the price index specified by the seller, bidders act as if they compete in a private values environment even if they are informed about future oil prices. Empirically, I fail to find evidence of adverse selection in bidding data from the 2022 drawdown, consistent with this implication of the model. If bidders are believed to be privately informed about local oil price levels, this suggests that index-based price adjustment is an effective auction design tool in the context of SPR drawdowns. Absent direct evidence of private information, which is difficult to measure in this context, I show that relatively little private information would be required in order for the impact of price indexing to be economically significant in an environment similar to the 2022 drawdown.

These results highlight the potential value of price indexing for sellers in settings where buyers’ valuations have important time-varying common components but goods cannot be delivered immediately. In particular, price indexing can be used to shape ex ante competition in a manner that benefits the seller. This perspective complements the classic long-term

²⁶If there is too much private information, equilibrium existence can fail. A similar phenomenon occurs in Vives (2010) and Vives (2011). See also Rostek and Yoon (2023).

contracting literature, which emphasizes the implications of price adjustment for contract performance within bilateral relationships (Goldberg, 1985; Goldberg and Erickson, 1987; Joskow, 1988). The apparently limited use of formal basis auctions in practice raises a question for future research.

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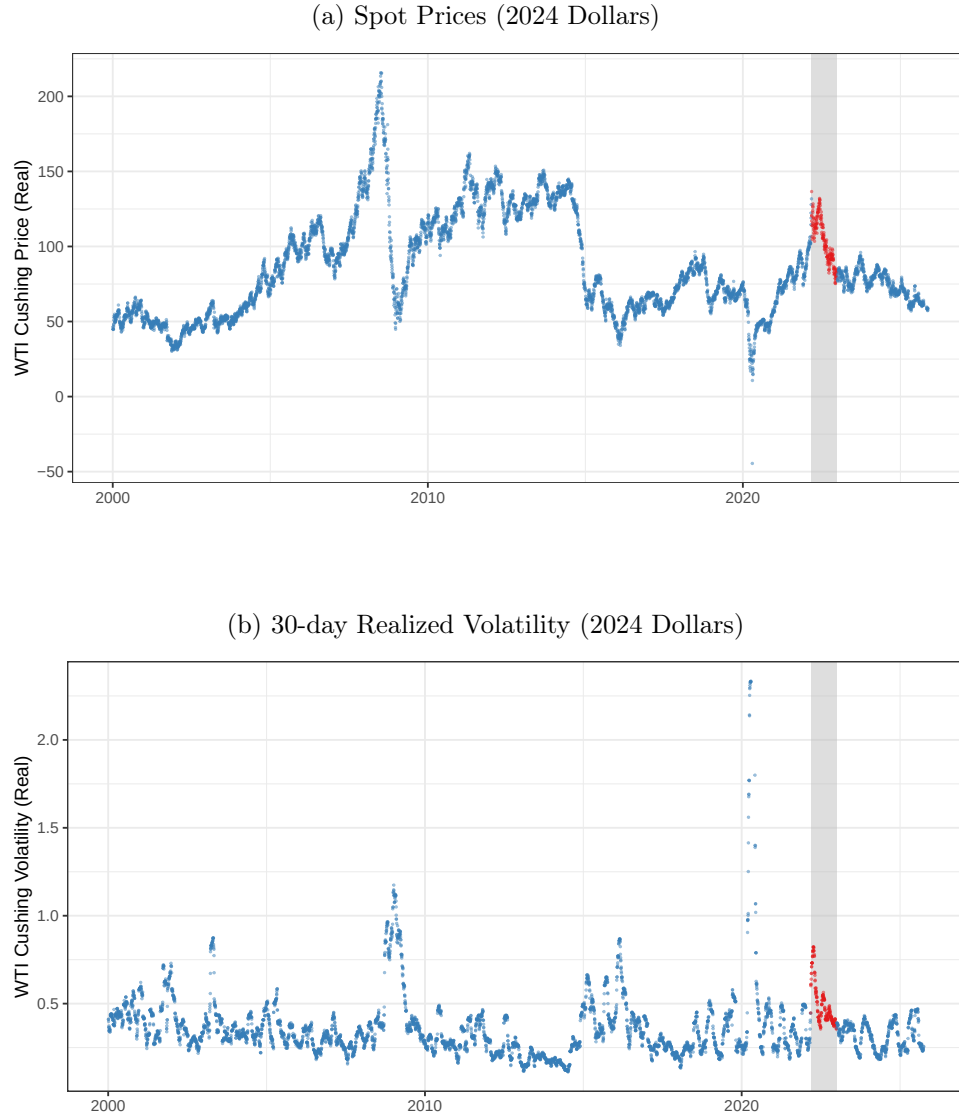
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Figure 3: WTI Cushing Spot Price and 30-Day Realized Volatility



Notes: (1) Spot price calculated as EIA Cushing, OK WTI Spot Price FOB price deflated by BLS Consumer Price Index for all goods less energy, all urban consumers, not seasonally adjusted (Series ID CUUR0000SA0LE); (2) Gray-shaded region spans March 8, 2022 (publication date of the first emergency sale Notice-of-Sale) through December 31, 2022 (end of delivery period for last emergency sale); (3) 30-day realized volatility calculated as the annualized 30-day standard deviation of the log price return. Sources: Bureau of Labor Statistics (2026), Bureau of Labor Statistics (2026); Energy Information Administration (2026).

Table 1: Mean Delivery Index Price (DIP) vs. Base Reference Price (BRP)

Date of Sale	Delivery Window	Sweet Crude			Sour Crude		
		BRP	DIP	% Δ	BRP	DIP	% Δ
2022-03-08	24-84 days	95.15	106.99	12.44	92.60	103.76	12.06
2022-04-12	33-79 days	111.00	116.57	5.02	106.28	108.34	1.93
2022-06-01	20-75 days	113.39	110.51	-2.54	108.81	94.54	-13.11
2022-06-28	49-94 days	121.42	89.15	-26.58	114.21	85.47	-25.16
2022-08-02	45-80 days	103.68	87.39	-15.71	97.61	82.77	-15.20
2022-09-27	35-64 days	89.45	87.56	-2.12			
2022-10-25	37-67 days	89.91	77.07	-14.28	83.86	71.39	-14.87

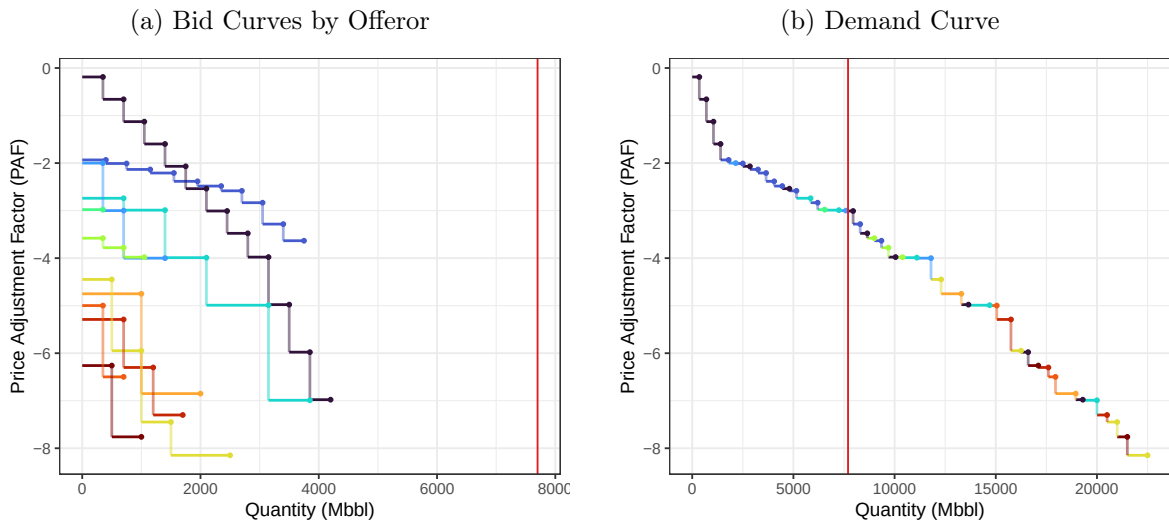
Notes: (1) Daily index price (DIP) is the mean daily closing price of Bloomberg tickers USCRMEHC (Crude Oil WTI Houston) and USCRMARS (Crude Oil Mars) during the Delivery Window. Delivery windows vary within sale by crude oil stream (MLI) and delivery location (DLI); the earliest and latest dates are used. (2) Base reference prices from Notice of Sale.

Table 2: Summary of Bidding Data (Sample Means)

	Sweet Crude			Sour Crude		
	Per DLI	Per DLI	Per MLI	Per DLI	Per DLI	Per MLI
	(Pipeline)	(Vessel)		(Pipeline)	(Vessel)	
Number of bidders	6.10	4.89	10.91	4.05	3.69	8.00
Number of bids	19.65	5.67	40.36	13.82	6.31	27.57
Number of winning bidders	3.30	1.11	6.00	2.36	1.23	4.36
Number of winning bids	8.55	1.25	16.45	7.71	3.20	13.86
Total quantity demanded	9906.75	2961.11	23962.27	7559.77	3880.38	16272.14
Total quantity sold	3825.00	643.75	7422.73	3927.14	1654.00	7072.14
Maximum winning PAF	0.42	-2.21	1.19	1.28	0.90	1.85
Average winning PAF	-0.76	-2.23	-0.93	0.23	0.06	-0.08
Minimum winning PAF	-2.03	-2.24	-2.89	-0.56	-0.58	-1.44
Nominal reserve PAF (5% BRP)	-5.12	-5.06	-5.13	-5.18	-5.21	-5.15

Notes: 2022 SPR drawdown bidding data. Includes all bids. (1) PAF refers to price adjustment factor; (2) Bid quantities are expressed in thousands of barrels (Mbbbl); (3) MLI refers to Master Line Item, designating the crude oil stream. DLI refers to Delivery Line Item, designating the place and mode of delivery. Crude oil from a single stream is typically available for delivery at 2-3 locations. Each bid must specify both a crude oil stream (MLI) and delivery location (DLI); (4) Nominal reserve calculated as -5% of the Notice of Sale Base Reference Price (BRP).

Figure 4: Example Bid and Demand Curves



Notes: (1) Bids for FY22 Emergency Drawdown No. 3B, West Hackberry Sweet Crude MLI, Pipeline DLI @ Sun (Non-SPR Tank). Each color corresponds to a different bidder. In this sale, the maximum and minimum DLI quantities coincide at 7,700 Mbbl, depicted by the vertical red line. (2) PAF refers to price adjustment factor; (3) Bid quantities are expressed in thousands of barrels (Mbbl).

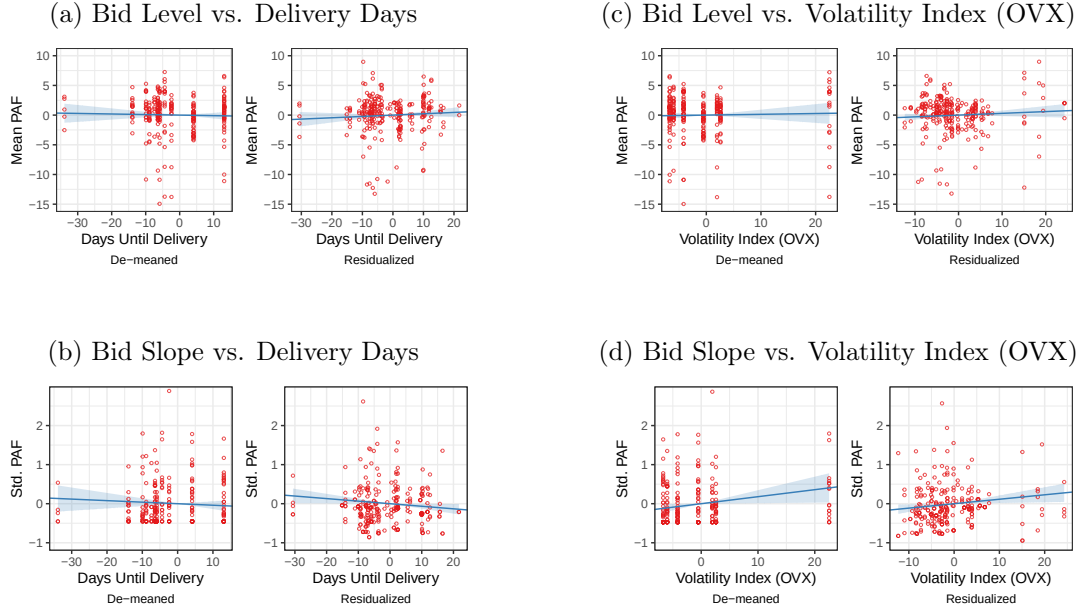
Table 3: Test for Private Information (Nyborg et al., 2002)

	<i>Dependent variable:</i>					
	Mean PAF		Std. PAF	Mean PAF		Std. PAF
	(1)	(2)	(3)	(4)	(5)	(6)
Volatility Index (OVX)	0.030 (0.024)	0.048 (0.037)	0.011** (0.005)			
Days Until Delivery				0.022 (0.020)	0.034 (0.024)	−0.007** (0.003)
Sweet Crude	−1.121** (0.492)	−1.111** (0.503)	0.003 (0.154)	−1.106** (0.479)	−1.072** (0.461)	0.025 (0.155)
Vessel DLI	−2.823*** (0.446)	−2.691*** (0.364)	0.091 (0.154)	−2.837*** (0.485)	−2.690*** (0.379)	0.109 (0.155)
Log Tranche Size (MMbbl)	−1.251*** (0.273)	−1.324*** (0.384)	0.249*** (0.043)	−1.210*** (0.272)	−1.262*** (0.338)	0.273*** (0.051)
Location FE	Yes	Yes	Yes	Yes	Yes	Yes
Dependent Variable	Level	Level	Slope	Level	Level	Slope
Unit of Observation	Bidder-Tranche	Tranche	Bidder-Tranche	Bidder-Tranche	Tranche	Bidder-Tranche
SE Clustering	Tranche	Event-Type	Tranche	Tranche	Event-Type	Tranche
Observations	268	54	268	268	54	268
Adjusted R ²	0.088	0.346	0.076	0.080	0.306	0.071

*p<0.1; **p<0.05; ***p<0.01

Notes: (1) Tranche refers to exclusive tranche, defined in Section 3.2. (2) Means and standard deviations of price adjustment factors (PAFs) are weighted by bid quantity; (3) Days Until Delivery is the number of days from the auction date to the midpoint of the delivery window; (3) Volatility Index (OVX) is the closing price of the CBOE Crude Oil Volatility Index on the auction date; (4) Observations are weighted by uncertainty (Days Until Delivery in (1)-(3), Volatility Index (OVX) in (4)-(6)); (5) Heteroskedasticity-robust standard errors clustered by sale-crude oil stream (MLI)-delivery location (DLI) for columns (1), (3), (4), (6) and sale-crude oil type (sweet/sour) for columns (2) and (5).

Figure 5: Correlations Between Bidding Strategy and Uncertainty



Notes: (1) Red circles indicate bidder-tranche quantity-weighted mean price adjustment factor (PAF) or standard deviation PAF. (2) For de-meaned panels, the regression line is obtained via bidder-tranche level regression weighted by uncertainty. (3) For residualized panels, the slope and regression line correspond to those reported in Table 3. The y- and x-axis variables are residualized on the independent variables in Table 3 (4) Confidence bands are heteroskedasticity-robust and clustered by sale-crude oil stream (MLI)-delivery location (DLI); (5) Days Until Delivery is the number of days from the auction date to the midpoint of the delivery window; (6) Volatility Index (OVX) is the closing price of the CBOE Crude Oil Volatility Index on the auction date.

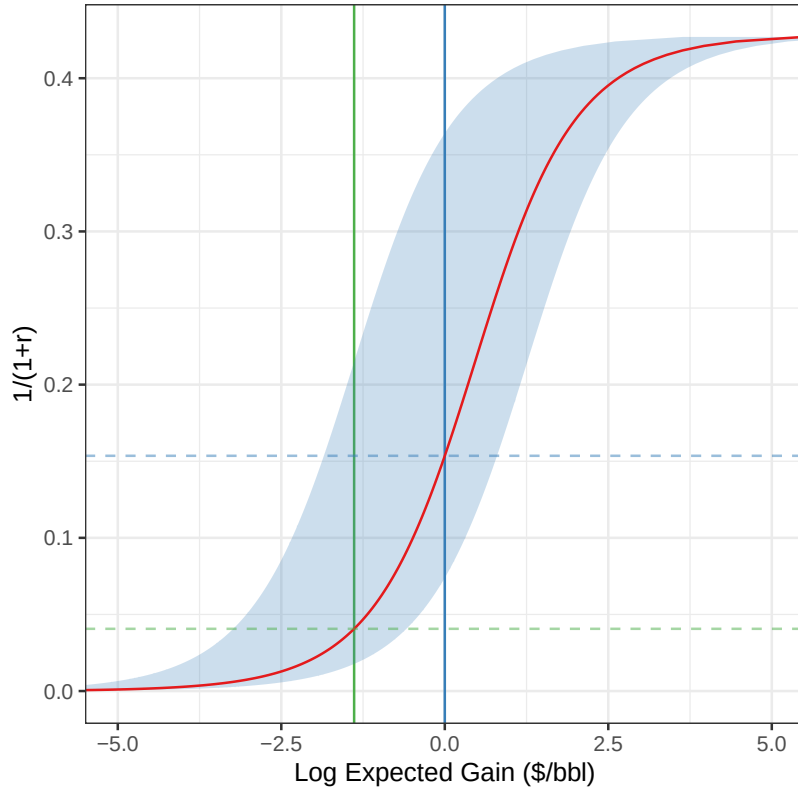
Table 4: Test of Independent Private Values (Hickman et al., 2021)

	(1)	(2)	(3)	(4)	(5)
Mean Rival PAF	0.074 (0.169)	0.008 (0.167)	-0.075 (0.135)	-0.075 (0.099)	-0.095 (0.131)
Days Until Delivery		0.010 (0.032)	0.005 (0.030)	0.005 (0.029)	0.002 (0.021)
Sweet Crude		-0.694 (0.546)	-0.172 (0.567)	-0.172 (0.670)	-0.688 (0.502)
Vessel DLI		-1.640*** (0.623)	-2.415*** (0.745)	-2.415*** (0.744)	-1.607*** (0.507)
Location FE	No	Yes	Yes	Yes	Yes
Polynomial in N	Cubic	None	Cubic	Cubic	Cubic
Restricted Sample	Yes	Yes	Yes	Yes	No
SE Clustering	Tranche	Tranche	Tranche	None	Tranche
Observations	144	144	144	144	217
R^2	0.070	0.090	0.115	0.115	0.101

*p<0.1; **p<0.05; ***p<0.01

Notes: (1) Tranche refers to exclusive tranche, defined in Section 3.2; (2) Excludes tranches not having at least two bidders willing to accept delivery location (DLI) minimum quantity; (3) Restricted sample excludes tranches with any bids stipulating minimum quantities greater than the Delivery Line Item (DLI) minimum quantity. Unrestricted sample retains all tranches but omits such bids from the sample and from Mean Rival PAF, but not from calculation of N ; (4) Days Until Delivery is the number of days from the auction date to the midpoint of the delivery window; (5) Heteroskedasticity-robust standard errors clustered by sale-crude oil stream (MLI)-DLI except for column (4), which is unclustered

Figure 6: Expected Revenue Gain from Price Indexing



Notes: (1) $1/(1+r) = \sigma_{\omega}^2/(\sigma_u^2 + \sigma_{\omega}^2)$ is the fraction of bidders' information attributable to private information about oil prices; (2) the blue (green) vertical line corresponds to a \$1/barrel (\$0.25/barrel) expected revenue gain from price-indexing; (3) 95% confidence band obtained via bootstrap estimation of the slope of bidder marginal valuations λ_{pred} .